Gradient Descent

Linear Regression (Recap)

- Goal:
  - Given a training set, learn θ
  - Such that $h_\theta(x)$ is as close as possible to y

Least Mean Squares (LMS)

- For a dataset with n samples, minimize

$$J(\theta) = \frac{1}{2}\sum_{k=1}^{n}\left(h_\theta(x^{(k)}) - y^{(k)}\right)^2$$

Cost Function/Loss Function

$$h_\theta(x) = \sum_{i=0}^{d}\theta_i x_i = \theta^T x$$

Linear Regression

- Goal: Choose θ to minimize $\mathrm{J}(\theta)$

$$\theta^* = argmin_\theta J(\theta)$$

- How?

- Gradient Descent

Gradient Descent

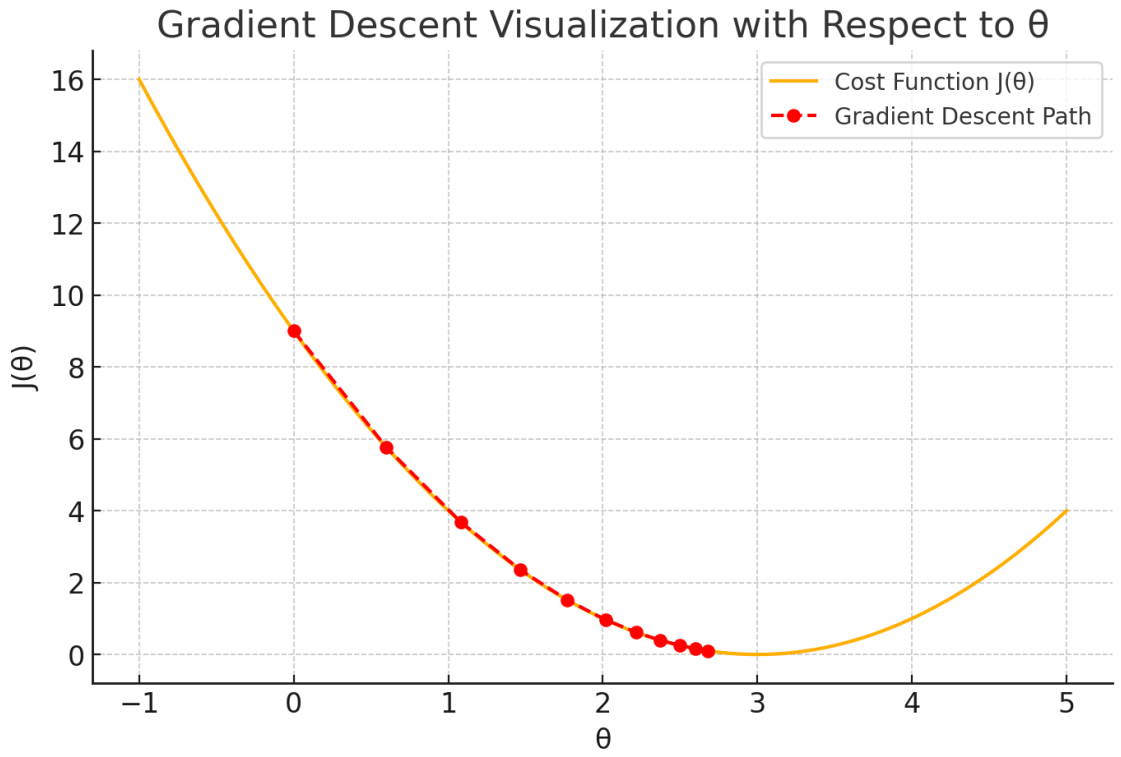
Cost Function

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(h_\theta(x^{(i)}) - y^{(i)})^2$$

- Start from an initial θ
- Repeatedly update θ as follows

$$\theta_j := \theta_j - \alpha\frac{\partial}{\partial\theta_j}J(\theta)$$

- θ_j corresponds to the coefficient of the x_j
- α is called learning rate (hyper-parameter)
  - Constant
  - Decaying



Gradient Descent

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

Cost Function

$$J(\theta) = \frac{1}{2} \sum_{i=1}^{n} (h_\theta(x^{(i)}) - y^{(i)})^2$$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{2} \sum_{k=1}^{n} \frac{\partial}{\partial \theta_j} \left(h_\theta(x^{(k)}) - y^{(k)}\right)^2$$

$$= \frac{1}{2} \sum_{k=1}^{n} 2\left(h_\theta(x^{(k)}) - y^{(k)}\right) \frac{\partial}{\partial \theta_j} h_\theta(x^{(k)})$$

$$= \sum_{k=1}^{n} \left(h_\theta(x^{(k)}) - y^{(k)}\right) \frac{\partial}{\partial \theta_j} \left(\sum_{i=0}^{d} \theta_i x_i^{(k)}\right)$$

$$= \sum_{k=1}^{n} \left(h_\theta(x^{(k)}) - y^{(k)}\right) x_j^{(k)}$$

Gradient Descent

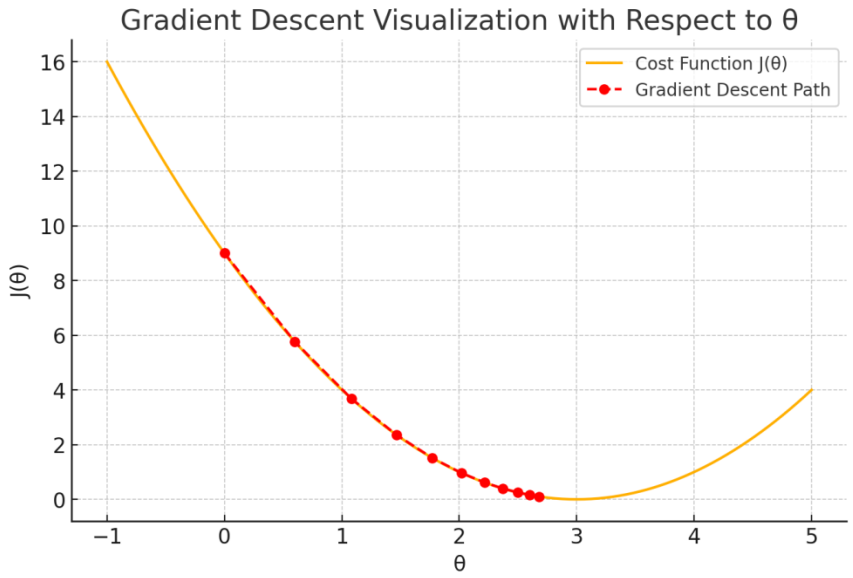
$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(h_\theta(x^{(i)}) - y^{(i)})^2$$

Cost Function

- Start from an initial θ

Repeat until convergence

$$\theta_j := \theta_j + \alpha \sum_{k=1}^{n} \left(y^{(k)} - h_\theta(x^{(k)})\right) x_j^{(k)}, \quad \text{(for every } j\text{)}$$



The Three Flavors of Gradient Descent

dataset এর sample কোনো কোনো সময় যাতে batch এ appear করে।

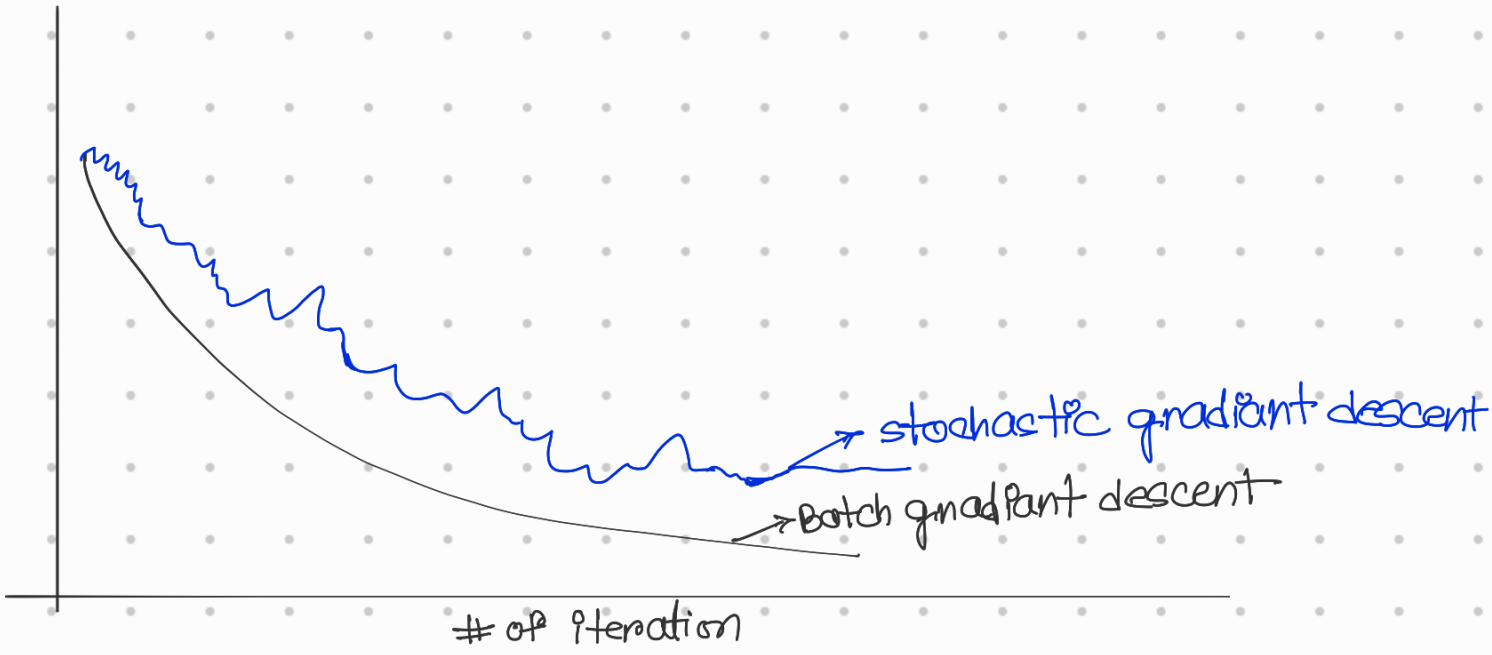
- Batch Gradient Descent
  - Use all n samples to compute the gradient
  - Exact gradient
  - Stable Updates
  - Slow, memory intensive

- Stochastic Gradient Descent
  - Use 1 sample to compute the gradient
  - Very fast updates
  - Noisy gradient, Convergence issues

→ mostly used

- Mini-batch Gradient Descent
  - Use a small batch of samples
  - Reasonably stable
  - Most used choice

আমাদের বুঝতে হবে on average gradient কত



(Randomly কয়েকটা sample নিয়ে তার mean বের করব, then batch gradient ??)

The Three Flavors of Gradient Descent

Let, $J(\theta, x, y)$ is the loss from sample (x, y)

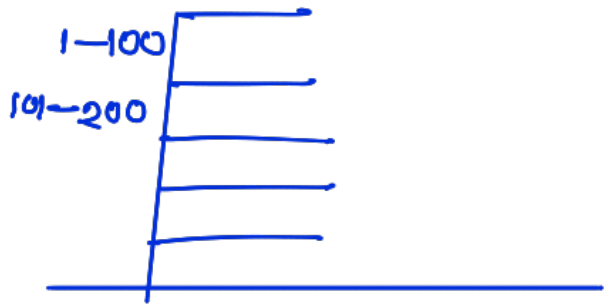
↳ random sample (x,y) এর উপর আমাদের loss.

- Batch Gradient Descent

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{N} \sum_{i=1}^{N} J(\theta, x_i, y_i)$$

- Stochastic Gradient Descent

$$\theta_j \leftarrow \theta_j - \alpha\, J(\theta, x, y)$$



- Mini-batch Gradient Descent

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} J(\theta, x_i, y_i)$$

$|\mathcal{B}|$ ⇐ # of Batch.

প্রতি Iteration এ আমরা all possible batch এর উপর Iterate করব।

Batch ⇒ পুরো dataset নিয়ে কাজ করা

MiniBatch ⇒ subset of dataset ,, ,,

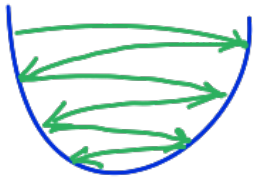
stochastic ⇒ random data set ,, ,,

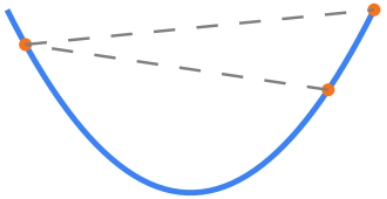
Learning Rate

- Too small

Painfully slow!

- Too Large

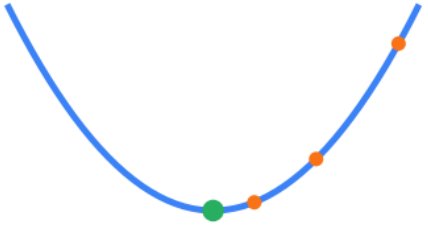




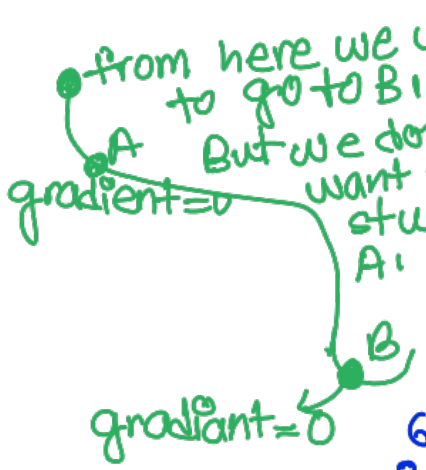
Oscillates/diverges!

We want avoid them

- Just Right



Converges nicely!



ভালো α বের করতে হবে।
Learning rate ↑
minima থেকে অনেকদূরে থাকলে
আমরা বড় বড় step নিয়ে minima তে
পৌঁছানোর try করি। কাছাকাছি আসলে ছোট ছোট
jump নিব, আবার বড় jump নিলে oscillating এর
বড়পাড়ি চলে আসবে।
(এটা পড়তে হবে??)

Reference

- D2l
- Deep Learning